

Anjali Singh, PhD

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Human-centered systems researcher and designer with 9 years of experience conducting research in human-AI interaction, learning, information seeking and human behavior. Led research across industry and academia to develop AI-powered systems for learning, search, and programming, through close collaboration with cross-functional teams. Experienced driving product and design decisions through empirical research, experimentation, behavioral analysis and stakeholder engagement.

Research Areas: Human-AI Interaction, Learning & Education, Information Seeking, Cognitive Science

Education

School of Information

Ph.D. in Information

University of Michigan, Ann Arbor

2019–2024

Mathematics and Computing

Integrated Bachelor and Master of Technology

Indian Institute of Technology, Delhi

2012–2017

Research and Work Experience

University of Texas at Austin | Postdoctoral Research Fellow

Enhancing Learning & Information Seeking with Generative AI

Sept'24–Current

- **Led the design, prototyping, and evaluation** of an **interactive GenAI-powered search and note-taking system**, that provides metacognitive support during information-seeking, helping users critically engage with information [\[Demo\]](#).
- **Developed novel behavioral metrics** including persistent inquiry and topic coverage using interaction logs (chat history, notes, and clickstream data) to **characterize user workflows** and **evaluate learning and exploration behaviors**.
- Designed & conducted Wizard-of-Oz studies, user interviews, and behavioral analyses to **understand how users search with GenAI, and interact with AI-generated guidance**, to inform design decisions for enhancing GenAI search.
- **Increased exploration breadth** by **88%**, **sustained inquiry behaviors** by **60%**, and **user questioning** by **46%**, compared to baseline system, demonstrating improved engagement and deeper information exploration.
- Applied **LLM-assisted qualitative coding, summarization and thematic analysis** to synthesize user data and outcomes and identify patterns in reasoning, trust, and engagement.

Responsible and Scalable AI-Integration in Higher Education

Sept'24–Current

- Conducted usability testing, interviews, and stakeholder research with instructors and administrators to identify user needs, product requirements and deployment considerations for a **campus-wide AI tutor platform at UT Austin**.
- Generated actionable **design recommendations spanning pedagogy, transparency, privacy, and user experience** that informed platform design and development.
- Designed & supported user studies evaluating impact of **multimodal GenAI tutors** on students' learning outcomes, engagement and experience, focusing on **trustworthiness, safety and accuracy**; Led to **0.61 σ boost in student scores**.
- Identified **opportunities for assessment redesign** in the AI era via instructor interviews, surveys, & co-design sessions.

Microsoft, Redmond + University of Michigan (U-M) | UX Research Intern

Understanding Data Science Learning and Feedback Workflows

Jan'22–July'22

- **Led design and deployment** of personalized feedback generation tool for data science courses at U-M, in partnership with 10-member cross-functional team of researchers, product managers, and developers across Microsoft and U-M.
- **Facilitated data sharing between U-M and Microsoft**. Identified common misconceptions and core competencies on which novice data scientists and learners require support through log analysis and qualitative code analysis.
- **Collected and analyzed user feedback**, including perceived helpfulness of feedback tool and level of learning support.
- **Disseminated findings with Microsoft leadership**, informing future directions for data science developer tools.

University of Michigan | Graduate Researcher

Learning Experience Design, Behavioral Analysis, Aligning with Student Motivations

Jan'20–Aug'24

- Designed and led **large-scale field experiments** involving **3800+ learners** to evaluate learning technologies promoting critical thinking, agency, and active learner-AI engagement, across informal and formal learning contexts.
- Generated **insights into motivation, participation, and effort** that informed the design of scalable learning systems.
- Developed **AI-integrated personalized learning experiences** using **LLM prompting** leveraging data from past cohorts; **Evaluated impact using metrics** such as retention, performance, learning outcomes and perceived value.
- Applied machine learning and behavioral clustering techniques to **identify patterns in user interaction and learning strategies using ebook log-data**, obtaining 3 distinct behavioral groups.

- *Visual Assessment Generation*. Generated insights for picture-based assessment development through interviews with instructors; Leveraged **image captioning, open information extraction, and neural embedding based semantic similarity matching** to auto-generate visual assessments using knowledge base and image corpus.
- *Civic Issue Reporting*. Conducted user research with Indian citizens to identify barriers and preferences for reporting civic issues. **Mined 1.5k images** with civic issues and associated captions for unsupervised training of deep learning model to automatically detect civic issues from images, **improving accessibility and effectiveness of civic reporting tools**.

Technical Skills & Interests

Quantitative Research Methods: A/B Testing & Experimentation, Statistical Modeling & Analysis (e.g., Regression, Cluster Analysis), Causal Inference, Behavioral Analysis, Log Analysis, Text Analysis, Applied AI/ML, Data Visualization

Qualitative Research Methods: Interviews, Surveys, User Studies, Usability Testing, Wizard-of-Oz studies, Contextual Inquiry, Focus Groups, Diary Studies, Co-design, Thematic Analysis

Technical: Python (NumPy, Pandas, SciPy, Matplotlib, Scikit-Learn, NLTK), R, SQL, React, Prompt Engineering, LLM APIs, AI-Assisted Prototyping

Tools & Platforms: Qualtrics, Prolific, Canva, Figma, Atlas.ti, MAXQDA, Jupyter, VSCode, RStudio, Github, Claude Code

Product & Collaboration: Research Strategy & Leadership, Stakeholder Engagement, Cross-functional Collaboration

Selected Awards

Best Paper Awards: ACM Learning@Scale Conference, 2021 | Learning Analytics and Knowledge Conference, 2024

Awarded \$200,000 Funding for Doctoral Research: Microsoft, 2022

Summer Undergraduate Research Award: Indian Institute of Technology Delhi, 2014

Gold Medal: Indian Physics Olympiad (InPhO), Awarded to Top 35 Indian high-school students as per InPhO exam, 2012

Selected Publications [[Google Scholar](#), 515+ citations, h-index: 9, 15+ publications]

Impact of Multimodal and Conversational AI on Learning Outcomes and Experience. K. Taneja, [A. Singh](#), A. K. Goel. *International Conference on Artificial Intelligence in Education (AIED)*. 2026 [[Paper](#)]

Toward Responsible and Scalable Integration of Course-Specific AI Tutors: Instructor Experiences with a Campus-Wide Platform. G. Ko, H. K. Lee, [A. Singh](#), L. Boddy, K. Ford, and E. Huff Jr. *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*. [[Paper](#)]

Enhancing Critical Thinking in Generative AI Search with Metacognitive Prompts. [A. Singh](#) et al. *Proceedings of the Association for Information Science and Technology (ASIS&T)*. 2025 [[Paper](#)]

What's In It for the Learners? Evidence from a Randomized Field Experiment on Learnersourcing Questions in a MOOC. [A. Singh](#). et al. *Proceedings of the Eighth ACM Conference on Learning@Scale (L@S)*. 2021. **Best Paper Award** [[Paper](#)]

Automatic Generation of Leveled Visual Assessments for Young Learners. [A. Singh](#) et al. *Proceedings of the AAAI Conference on Artificial Intelligence*. 2019. [[Paper](#)]

Research Outreach & Community Leadership

Invited Talk: Designing Technologies for Critical and Active Learning in the Age of AI. *EPFL*, 2025

Organised Workshop: Tools for Thought: Understanding, Protecting, and Augmenting Human Cognition with Generative AI—From Vision to Implementation. *CHI 2026 Conference*

Organised Workshop: Empowering Education with LLMs - the Next-Gen Interface and Content Generation. *Artificial Intelligence in Education Conference*, 2023

Invited Talk: Feedback Generation using Human-AI Collaboration for Introductory Data Science. *Microsoft*, 2021

Leadership & Mentoring

2020–Current: Program Committee Member/Reviewer. *Conferences:* CHI ('25,'24,'23), Learning@Scale ('25,'26), Learning Analytics and Knowledge'26, SIGCSE ('24,'23), Educational Data Mining Conference ('21); *Journals:* Computers & Education, Human-Computer Interaction, Journal of Learning Analytics

2018–Current: Mentored 5 undergraduate and 6 graduate students at the University of Michigan, UT Austin, and Georgia Tech and 3 research fellows and interns at IBM Research and Microsoft

2021: Mentored 2 teams in Educational Data Mining track, **LearnLab Summer School, Carnegie Mellon University**

2018: Organising team member for **Hackathon on AI for Social Good** at IBM Research Labs India